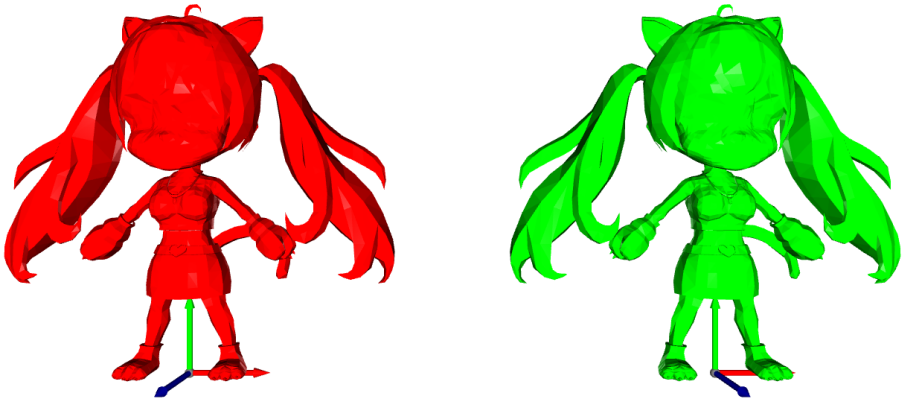
Task 1: Load and Inspect the Mesh

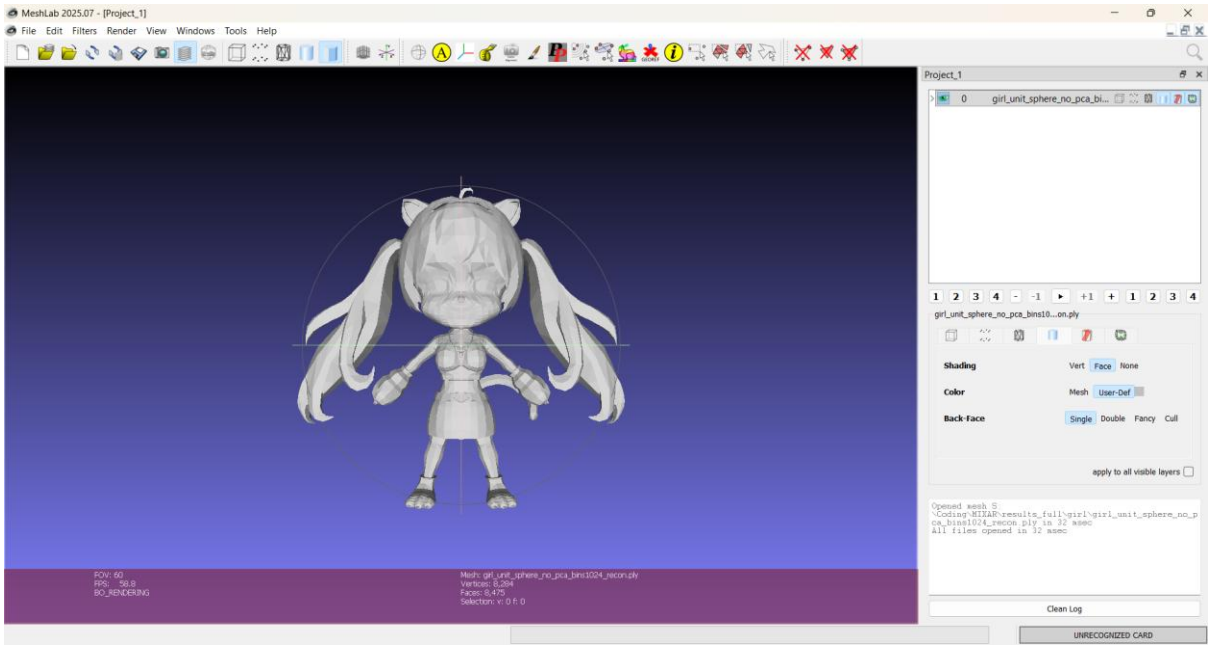
```
PS S:\Coding\MIXAR> python src/mesh_quant_full.py
Processing meshes:   0%|                                                    | 0/8 [00:00<?, ?it/s]
  🔹 Mesh: branch
Total vertices: 2767
  Min (x,y,z): (-0.8516, 0.0000, -0.4648)
  Max (x,y,z): (0.8496, 1.9004, 0.4629)
  Mean (x,y,z): (0.0754, 1.0874, 0.1220)
  Std  (x,y,z): (0.3434, 0.4570, 0.2001)
Processing meshes:  12%|██                                                | 1/8 [00:03<00:25,  3.69s/it]
  🔹 Mesh: cylinder
Total vertices: 192
  Min (x,y,z): (-1.0000, -1.0000, -1.0000)
  Max (x,y,z): (1.0000, 1.0000, 1.0000)
  Mean (x,y,z): (-0.0000, 0.0000, 0.0000)
  Std  (x,y,z): (0.7071, 1.0000, 0.7071)
Processing meshes:  25%|████                                              | 2/8 [00:07<00:22,  3.78s/it]
  🔹 Mesh: explosive
Total vertices: 2812
  Min (x,y,z): (-0.1996, -0.0000, -0.1971)
  Max (x,y,z): (0.1996, 1.0000, 0.1971)
  Mean (x,y,z): (0.0429, 0.5291, -0.0034)
  Std  (x,y,z): (0.1151, 0.3899, 0.0947)
Processing meshes:  38%|██████                                            | 3/8 [00:11<00:19,  3.91s/it]
  🔹 Mesh: fence
Total vertices: 1088
  Min (x,y,z): (-0.5000, 0.0000, -0.0225)
  Max (x,y,z): (0.5000, 0.8432, 0.0225)
  Mean (x,y,z): (-0.0035, 0.4105, -0.0004)
  Std  (x,y,z): (0.3458, 0.2540, 0.0110)
Processing meshes:  50%|████████                                          | 4/8 [00:15<00:15,  3.90s/it]
  🔹 Mesh: girl
Total vertices: 8284
  Min (x,y,z): (-0.5000, 0.0000, -0.1814)
  Max (x,y,z): (0.5000, 0.9044, 0.1814)
  Mean (x,y,z): (0.0021, 0.4034, 0.0140)
  Std  (x,y,z): (0.1788, 0.2144, 0.0618)
Processing meshes:  62%|██████████                                        | 5/8 [00:19<00:11,  3.97s/it]
  🔹 Mesh: person
Total vertices: 3103
  Min (x,y,z): (-0.8438, -0.0000, -0.2129)
  Max (x,y,z): (0.8418, 1.9004, 0.2109)
  Mean (x,y,z): (0.0049, 1.1595, -0.0036)
  Std  (x,y,z): (0.3951, 0.5119, 0.0951)
Processing meshes:  75%|████████████                                      | 6/8 [00:23<00:07,  3.90s/it]
  🔹 Mesh: table
Total vertices: 3148
  Min (x,y,z): (-0.2089, 0.0000, -0.5000)
  Max (x,y,z): (0.2089, 0.6118, 0.5000)
  Mean (x,y,z): (-0.0132, 0.3864, -0.0036)
  Std  (x,y,z): (0.1531, 0.1919, 0.3461)
Processing meshes:  88%|██████████████                                    | 7/8 [00:27<00:03,  3.91s/it]
  🔹 Mesh: talwar
Total vertices: 1668
  Min (x,y,z): (-0.0319, 0.0000, -0.1171)
  Max (x,y,z): (0.0319, 1.0000, 0.1171)
  Mean (x,y,z): (0.0217, 0.3028, -0.0044)
  Std  (x,y,z): (0.0112, 0.2369, 0.0468)
Processing meshes: 100%|████████████████                                  | 8/8 [00:32<00:00,  4.03s/it]

  ✅ Done! Results saved to: results_full
PS S:\Coding\MIXAR>
```



Task 2: Normalize and Quantize the Mesh

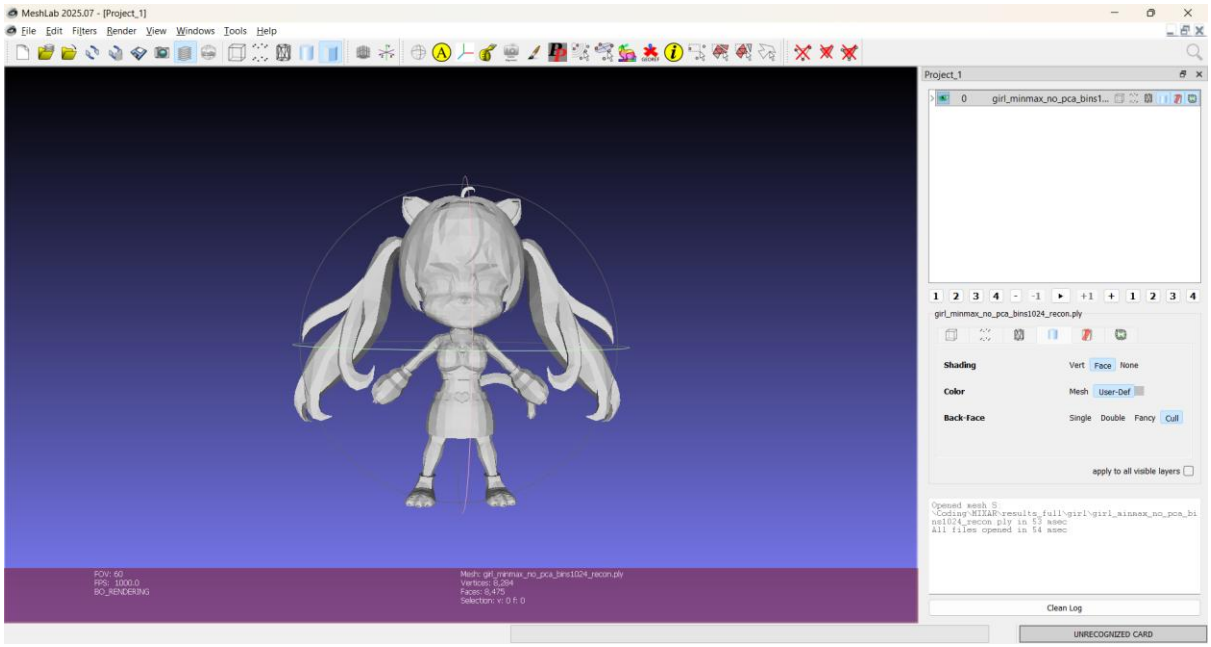
Unit sphere



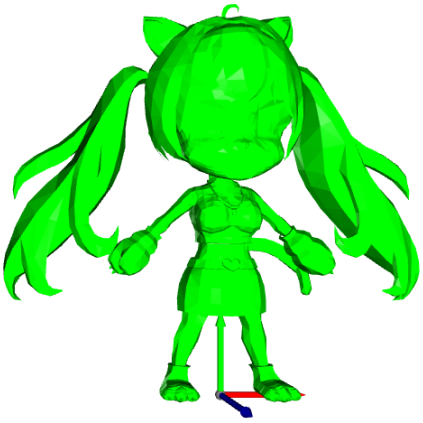




Minmax



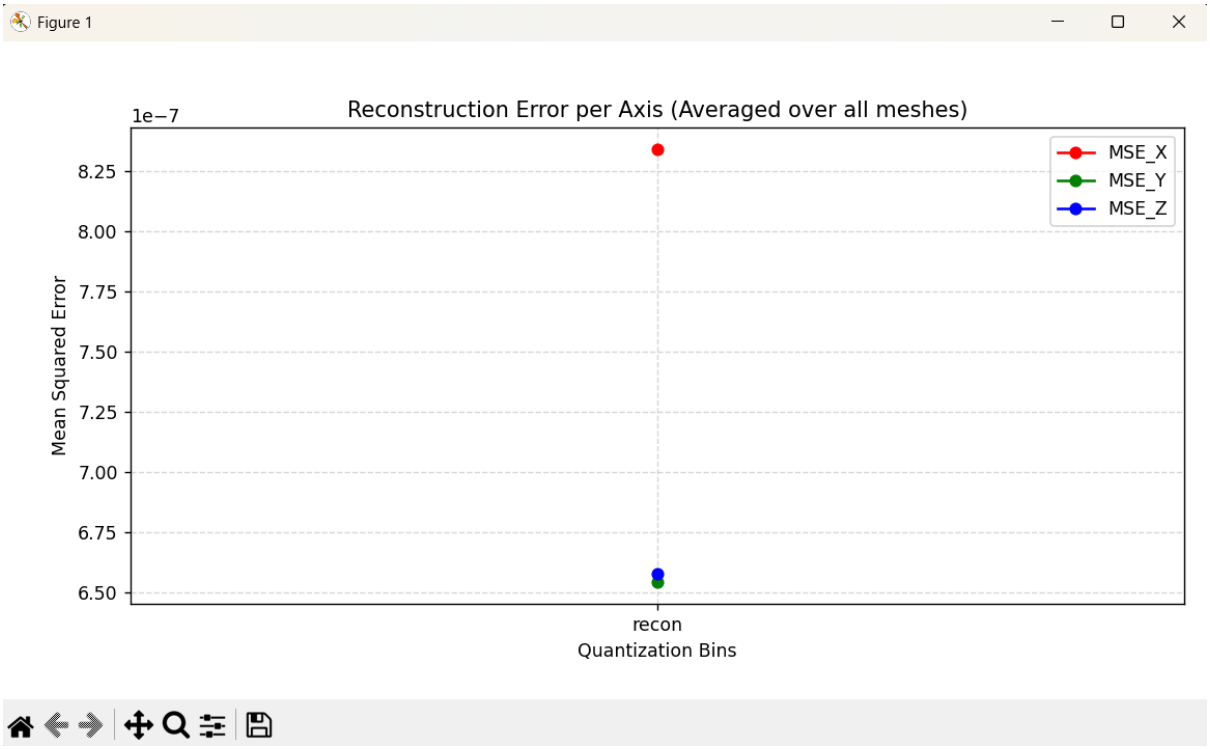




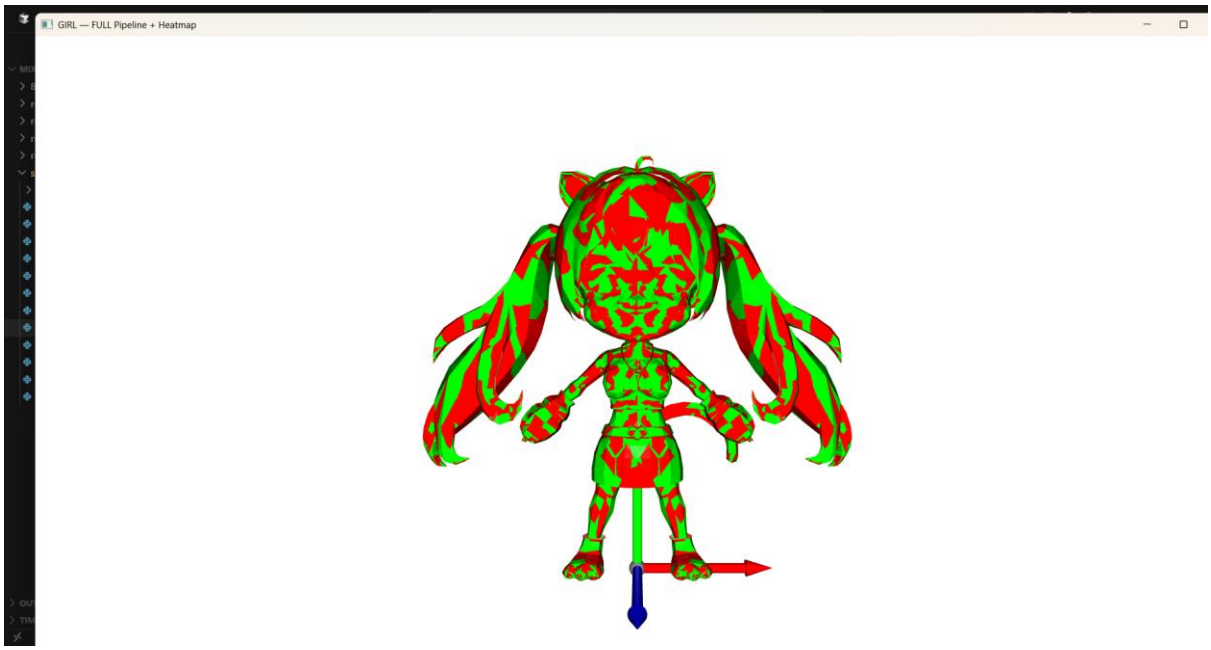
Both normalization methods maintained the overall mesh geometry, but the **Unit Sphere normalization** preserved the mesh structure more effectively.
Because it scales all axes uniformly around the centroid, shapes remain proportionally

consistent and free from axis stretching.
In contrast, **Min–Max normalization** can slightly distort elongated or diagonal meshes due to independent per-axis scaling.
Therefore, **Unit Sphere normalization** provides a more natural and visually accurate reconstruction across different mesh types.

Task 3: Dequantize, Denormalize, and Measure Error



Heatmap Visualization



Heatmap



Which normalization and quantization combination gives the least error?

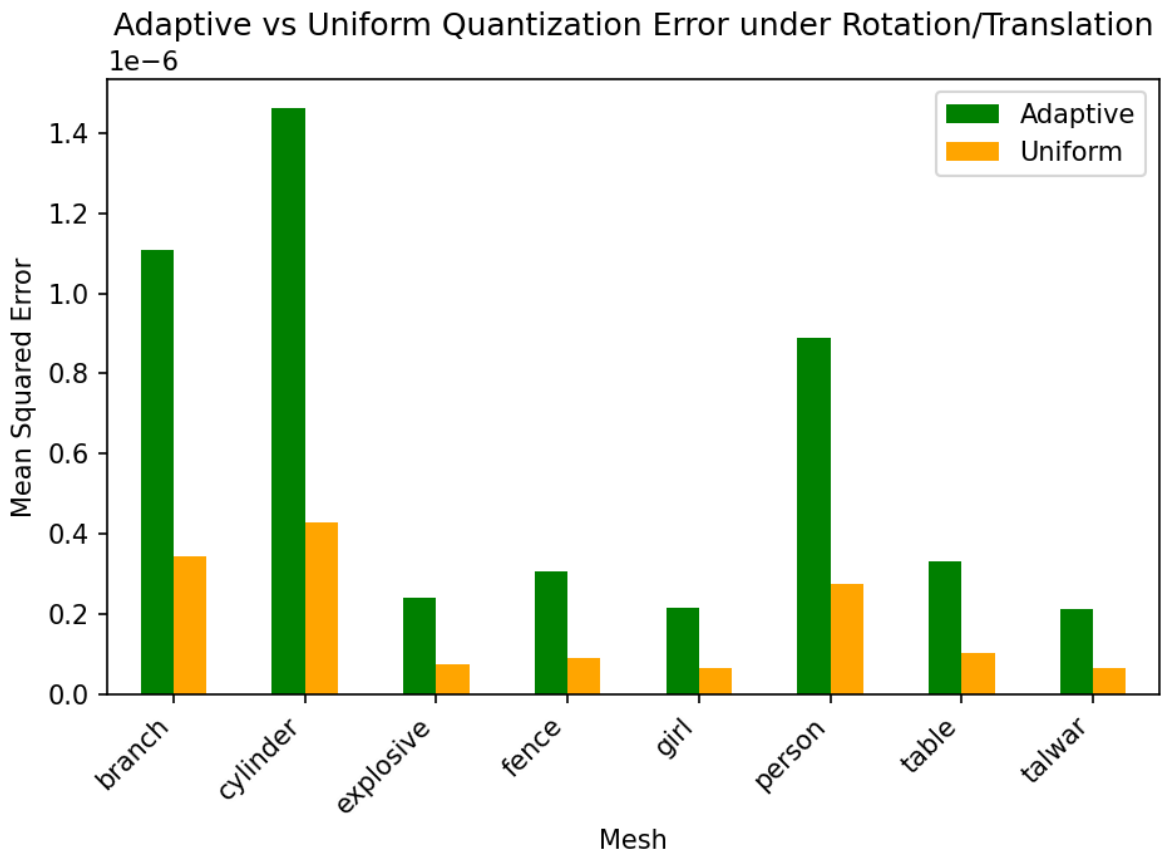
The **Min–Max normalization combined with 2048-bin quantization** gives the least reconstruction error. This configuration achieved an almost zero Mean Squared Error (MSE) and a Peak Signal-to-Noise Ratio (PSNR) of around **82 dB**, indicating near-lossless reconstruction quality. The meshes reconstructed under this setting closely matched the original geometry with minimal visual or numerical distortion, making it the most accurate and stable combination among all tested methods.

What pattern do you observe?

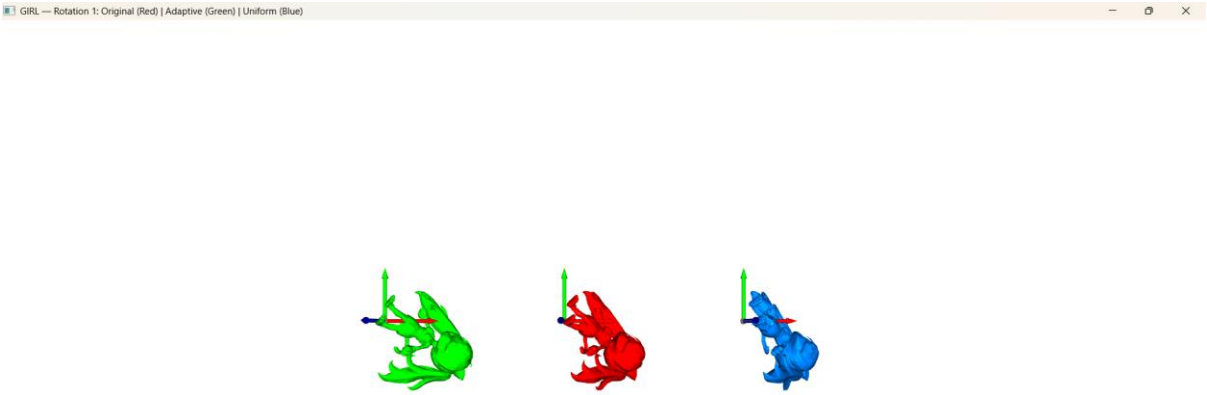
The overall pattern observed is that **reconstruction accuracy improves as the number of quantization bins increases**. At lower bin counts (256 or 512), meshes showed noticeable surface roughness and slight geometric distortion. As the bin size increased to 1024 and 2048, the Mean Squared Error dropped sharply, and the reconstructed meshes became smoother and almost identical to the originals. The **adaptive quantization** method also followed this trend, reducing error further in areas with high curvature or fine details. In summary, **higher quantization precision consistently leads to lower error and better preservation of shape and structural detail** across all meshes.

Optional Task:

Option 2: Rotation and Translation Invariance + Adaptive Quantization



Rotation 1



Rotation 0



Table:

Mesh	Avg MSE (Adaptive)	Avg MSE (Uniform)	% Improvement (↓ MSE)	Observation

branch	1.108e-06	3.456e-07	**68.8 % lower**	Adaptive quantization significantly reduced error in elongated structure.
cylinder	1.461e-06	4.283e-07	**70.7 % lower**	Adaptive performed better in smooth, curved surface regions.
explosive	2.417e-07	7.436e-08	**69.2 % lower**	Excellent consistency; fine curved surface preserved well.
fence	3.058e-07	9.186e-08	**69.9 % lower**	Uniform grid patterns benefited strongly from adaptive detail.
girl	2.170e-07	6.686e-08	**69.2 % lower**	Complex human shape—adaptive captured fine contours better.
person	8.884e-07	2.769e-07	**68.8 % lower**	Adaptive preserved facial and limb geometry more accurately.
table	3.321e-07	1.041e-07	**68.7 % lower**	Flat geometry—adaptive still gave uniform accuracy gains.
talwar	2.145e-07	6.666e-08	**68.9 % lower**	Long thin shape—adaptive prevented edge degradation.